

Reconstructing Axially Symmetric 3D Pottery Using Genetic Algorithms

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Abstract—The computational reassembly of broken archaeological pottery is a combinatorial challenge due to the large search space and geometric ambiguity from thin fracture surfaces. Existing methods rely on greedy incremental search or data-driven learning, each with limitations in scalability or generalization. We present a genetic algorithm (GA) system for 3D pottery reassembly, targeting the reconstruction of individual axially symmetric pots from their constituent sherds. Building on the Structure-from-Sherds++ (SfS++) pipeline, we replace the BAISER beam search with a population-based evolutionary search while preserving the geometric feature extraction and pairwise matching stages. Our GA uses an integer chromosome representation over precomputed match candidates, a shard-partition crossover operator, shard-centric mutation, and a multi-component fitness function combining match quality, cycle consistency, edge and rotation residual penalties, connectivity rewards, and voxel-based collision detection. Experiments on the SfS++ benchmark dataset show that our approach achieves results comparable to SfS++ on some pots while lagging on others, showing the feasibility of evolutionary search and identifying areas for improvement. We further present convergence analysis and visual reconstruction progression to show how the evolutionary search builds up assemblies over time.

Index Terms—genetic algorithm, pottery reassembly, evolutionary computation, 3D reconstruction, archaeological fragment assembly, fitness function design, combinatorial optimization

I. INTRODUCTION

The reassembly of broken pottery recovered from archaeological sites remains a longstanding challenge in archaeology. Studying these artifacts is essential for understanding the societies that once inhabited these sites. However, even skilled archaeologists and restoration experts with knowledge of historical context and geometric structure face a tedious and time-consuming process, often requiring days to reconstruct a single pottery [1].

This raises the need for computer-driven methods that can solve this problem more efficiently and in significantly less time. By nature, this is a combinatorial problem: even a single pottery broken into 10 pieces can have 3,628,800 possible arrangements, not accounting for the rotational and translational degrees of freedom of each fragment. Over the past few years, significant research has been dedicated to this task, one branch of which focuses on axially symmetric potteries [2].

Existing approaches to this problem range from geometric feature matching to deep learning and parallel search methods,

each with limitations in scalability, data requirements, or computational cost [1]. Among the most successful recent methods is Structure-from-Sherds++ (SfS++) [2], which draws inspiration from incremental Structure-from-Motion pipelines. SfS++ extracts axis-based geometric features and breakline descriptors from each sherd, performs pairwise matching using the Longest Common Subsequence (LCS) algorithm, and then assembles the pot incrementally using a multi-graph beam search technique called BAISER (Base-Agnostic Incremental ShErd Registration). SfS++ achieves state-of-the-art results on real pottery datasets, including the task of simultaneously reconstructing multiple pots from mixed fragment collections.

In this paper, we focus on reconstructing individual axially symmetric pots where all sherds belong to the same vessel. We build on the SfS++ pipeline, preserving its feature extraction and pairwise matching stages, and replace the BAISER beam search with a genetic algorithm. We present a GA-based solution for 3D pottery reassembly and analyze its reconstruction accuracy and computational cost.

II. LITERATURE REVIEW

A. Geometric Feature Matching Approaches

A standard computational pipeline for reassembly begins with surface segmentation, where algorithms distinguish a sherd's intact exterior from its fractured surfaces using either intrinsic geometric properties or learning-based techniques. Following segmentation, pairwise matching methods utilize geometric descriptors or boundary curves extracted from fracture surfaces to estimate the relative rigid alignment between complementary sherds [1], [3].

Huang et al. [4] present a system for automatic digital reassembly of broken 3D objects using only the geometric properties of fragments. Unlike earlier methods that required nearly flat pieces or perfect matches, their pipeline handles fragments with arbitrary, irregular geometries and partial matches. Notably, they introduce a multi-piece matching approach that aligns all fragments simultaneously, preventing the compounding errors that arise when pieces are joined one at a time.

Winkelbach and Wahl [5] propose a surface matching method for reassembling broken 3D solids. They use a hierarchical cluster tree of oriented points rather than relying on surface features or requiring an initial pose estimate. The method

systematically searches the full space of possible contact poses through a simultaneous depth-first traversal of both fragments' cluster trees. The method further prunes hypotheses whose upper-bound contact quality falls below the best match found so far. The approach guarantees termination and returns the globally optimal pose, achieving accurate alignments in under nine seconds across diverse test cases. However, it addresses only pairwise matching and does not scale to the full multi-fragment reassembly problem.

Papaioannou et al. [6] extend this direction by combining fracture surface matching with salient features extracted from the intact sides of fragments, making the approach more robust when fracture surfaces alone are insufficient for reliable alignment. Their pipeline further incorporates a symmetry-based completion step to recover missing geometry, producing a more complete restoration. However, approaches that rely purely on geometric matching still struggle in the presence of noise, missing fragments, and false positive matches, limiting their effectiveness in real-world scenarios [1].

Gregor et al. [7] propose a broader automated pipeline for reconstructing defective cultural heritage artifacts. Their pipeline classifies surfaces to distinguish intact areas from fractured edges using Heat Kernel Signatures, a geometric descriptor robust to damage and deformation, before performing reassembly, symmetry-based completion, and inpainting in sequence. For axially symmetric pottery in particular, rotational symmetry plays a critical role in guiding reassembly by heavily constraining the search space [1]. However, their pipeline is designed as a general cultural heritage restoration tool rather than one tailored to pottery reassembly specifically.

Pintus et al. [8] survey geometric analysis techniques in cultural heritage, organizing the field by geometric scale and object cardinality. They identify pottery reconstruction as a domain that benefits from axial symmetry. They further note that the multi-fragment assembly problem, framed as a scalable and globally optimal combinatorial search, remains an open challenge. Their taxonomy shows that no existing approach combines population-based search with domain-specific symmetry constraints for 3D pottery.

B. Data-Driven and Deep Learning Approaches

Some approaches use deep learning techniques to train convolutional neural networks (CNNs) on the fragment's geometry. These data-driven methods require significant amounts of well-labeled training data and long training times, posing practical challenges in archaeological settings where labeled data is scarce [1].

Pinho et al. [9] propose PotNet, a 1D convolutional neural network that predicts the rotation and translation required to place a sherd's point cloud into its correct position within the original vessel's coordinate system. To address the scarcity of real-world training data, the method relies on synthetically generated fragments produced by virtually shattering complete vessel models. Evaluated on both synthetic fragments and real-world scanned sherds, PotNet achieves average positioning errors in the centimeter range with millimeter-level standard

deviations. Rather than producing a final reconstruction directly, it narrows the search space by providing a strong initial position estimate, which is subsequently refined by classical algorithms such as Iterative Closest Point (ICP). While promising, this approach inherits the general limitations of data-driven methods: its accuracy depends on the quality and diversity of synthetic training data, and generalization to unseen vessel shapes or fragment types remains a challenge.

C. Parallel Search Approaches

Other approaches rely on parallelism, constructing a search tree with multiple branches explored simultaneously to find the best possible reconstruction. While thorough, these methods require substantial computational resources and significant runtime [1]. A notable example is the BAISER beam search technique employed by Yoo et al. [2], which focuses on axially symmetric pottery and is the foundation of our work.

D. Genetic Algorithms for Reconstruction

Genetic algorithms have been successfully applied to 2D reconstruction problems, demonstrating their ability to navigate large combinatorial search spaces without committing to early greedy decisions.

Sholomon et al. [10] propose the first effective genetic algorithm-based jigsaw puzzle solver. Their core contribution is a crossover procedure that merges two parent solutions into an improved child by detecting, extracting, and combining correctly assembled segments, an approach grounded in the best-buddy similarity metric, which identifies pairs of puzzle pieces that are mutually the best match for each other along a given edge. Their solver achieved state-of-the-art performance on puzzles of sizes previously unattempted.

Building on this, Sholomon et al. [11] extend their solver to handle puzzles with unknown dimensions and unknown piece orientation, successfully reconstructing a puzzle of 22,755 pieces. Their solver also shows the ability to simultaneously reconstruct a mixed bag of 16,405 pieces drawn from seven different images, showing the scalability of genetic approaches to large and ambiguous reassembly problems.

Sizikova and Funkhouser [12] address the global assembly of 2D fragmented wall paintings suffering from missing pieces and severe erosion. Traditional greedy methods fail on such artifacts because early incorrect matches propagate errors irreversibly. Their unsupervised genetic algorithm instead maintains a diverse population of partial reconstructions, evaluating clusters via a fitness function that rewards dense match connections and geometrically sound convex hulls. This work is of relevance to our approach, as it shows the feasibility of genetic algorithms for noisy, real-world fragment reassembly.

To our knowledge, no prior work has applied genetic algorithms to 3D pottery reassembly, leaving this a largely unexplored direction that we address in this paper.

III. METHODOLOGY

A. Problem Formulation

The problem consists of two subproblems: (i) geometric feature extraction and matching of sherds, and (ii) the reassembly process itself [1].

Let $\mathcal{S} = \{s_1, s_2, \dots, s_n\}$ denote the set of n pottery sherds, each represented as a 3D point cloud with associated surface normals. The goal is to find a set of rigid transforms $\{T_1, T_2, \dots, T_n\}$ and a connection graph G over $\mathcal{S}$ such that the assembled configuration maximally reconstructs the original vessel. Each edge (i, j) in G represents a physical contact between sherds s_i and s_j , with an associated relative transform T_{ij} .

Following SfS++ [2], we restrict our focus to axially symmetric pottery, where rotational symmetry about the vertical axis substantially constrains the search space.

B. Pipeline Overview

Our system utilizes the geometric preprocessing stages of SfS++ and applies a Genetic Algorithm (GA) for the global assembly. The full pipeline is:

- 1) **Feature extraction:** Axis estimation, breakline extraction, and rim segmentation per sherd (using SfS++).
- 2) **Pairwise matching:** LCS-based matching produces a candidate pool $\mathcal{M}$ of pairwise match hypotheses (using SfS++).
- 3) **Pairwise pruning:** Geometric consistency filtering reduces $|\mathcal{M}|$ to a tractable candidate set (using SfS++).
- 4) **Pair group construction:** Candidates in $\mathcal{M}$ are grouped by shard pair, forming $\mathcal{G} = \{G_1, G_2, \dots, G_m\}$ where each G_k contains all match hypotheses between a specific pair of sherds.
- 5) **GA search:** A population-based search over chromosomes encoding which pair groups to accept and which match hypothesis to select in the accepted pair groups.

In our approach, the geometric feature extraction and pairwise candidate generation are entirely handled by the SfS++ system. Our primary contribution is the combinatorial global reassembly process via evolutionary optimization.

C. Chromosome Representation

A chromosome $\mathbf{c}$ is an integer vector of length $m = |\mathcal{G}|$, one gene per pair group. Gene $c_k \in \{0, 1, \dots, |G_k|\}$ encodes the match selection decision for pair group G_k :

$$c_k = \begin{cases} 0 & \text{reject all matches for this pair} \\ j & \text{accept the } j\text{-th match hypothesis in } G_k \end{cases} \quad (1)$$

D. Population Initialization

Each chromosome is initialized into a highly dense state. For every available pair group G_k that contains candidate matches, the algorithm actively assigns an initial candidate $j \geq 1$, effectively turning on every possible edge in the connection graph.

To select these initial candidates, they are assigned a density score:

$$\text{density}(j) = \frac{\text{inlier}(j)}{1 + \text{score}(j)} \quad (2)$$

where $\text{inlier}(j)$ represents the number of breakline correspondences falling within a distance threshold during registration, and $\text{score}(j)$ is the LCS dissimilarity metric.

We employ a weight-based roulette wheel selection mechanism to sample the candidates. The probability of selecting candidate j in group G_k is strictly proportional to its density score:

$$P(\text{select } j) = \frac{\text{density}(j)}{\sum_{j' \in G_k} \text{density}(j')} \quad (3)$$

This policy ensures the initial dense graph is constructed using geometrically reliable match hypotheses while still allowing for broad exploration.

E. Chromosome Repair

The initialization (and subsequent mutation operations) produce densely connected graphs filled with physical cycles. However, to reconstruct a single pot composed of N sherds without internal structural cycles, a valid assembly mathematically requires exactly $N - 1$ active edges, forming a tree.

We resolve this by applying a chromosome repair function prior to fitness evaluation. The repair algorithm evaluates the massive pool of active edges generated by the chromosome, sorts them by their geometric density scores, and applies Kruskal's algorithm via a Disjoint Set Union (DSU). This process prunes the active edges down to exactly the best $N - 1$ pairs, forcing all remaining genes to 0. As a result, the repaired chromosome always represents a valid maximum spanning tree.

F. Fitness Function

The fitness function $F(\mathbf{c})$ evaluates the geometric quality of the assembly encoded by the repaired chromosome $\mathbf{c}$. The function aggregates four primary components to compute the total fitness:

$$F(\mathbf{c}) = F_{\text{inlier}} + F_{\text{consensus}} - F_{\text{overlap}} - F_{\text{comp}} \quad (4)$$

1) **Inlier Reward (F_{inlier}):** For each active match $c_k > 0$, the algorithm computes an inlier reward equal to the density metric of the selected candidate. This serves as the foundational geometric pull of the fitness function:

$$F_{\text{inlier}} = \sum_{k | c_k > 0} \text{density}(c_k) \quad (5)$$

2) **Consensus Reward ($F_{\text{consensus}}$):** The consensus reward assesses configurations where multiple sherds structurally agree on a global transform, forming stable local clusters. For every pair of sherds (i, j) in the assembly that are *not* connected by an active gene, the algorithm searches their pair group G_{ij} for any currently inactive candidate transform $T_{\text{candidate}}$ that mathematically agrees with their assembled relative transformation $T_j \cdot T_i^{-1}$. If the translation and rotation residuals between this inactive candidate and the actual

assembled configuration fall within strict thresholds (τ_t and τ_R), a consensus reward is accumulated proportional to that supporting candidate’s density:

$$F_{\text{consensus}} = \lambda_{\text{cons}} \sum_{(i,j) \notin \text{active}} \mathbb{I}(\text{consensus_match}(i,j)) \cdot \text{density}(j) \quad (6)$$

This rewards assemblies that are mutually supported by multiple unused candidate matches, effectively validating the configuration through implicit loop closures.

3) *Overlap Penalty* (F_{overlap}): To penalize physically implausible interpenetration, we project each shard into the assembled global space. We construct voxel-downsampled collision clouds (using a 3.0 mm grid resolution) for each shard, excluding points within a narrow margin of the breakline edges to avoid penalizing valid contacts. For any pair of non-adjacent sherds (i, j) in the assembly, we use kd-tree nearest-neighbor queries to compute a hit ratio:

$$h_{ij} = \frac{|\{p \in \mathcal{P}_i \mid \text{dist}(p, \mathcal{P}_j) < \epsilon_{\text{coll}}\}|}{|\mathcal{P}_i|} \quad (7)$$

If the accumulated hit ratio for a shard exceeds a minimum threshold (h_{min}), a penalty is applied:

$$F_{\text{overlap}} = \lambda_{\text{overlap}} \sum_i \max(0, \sum_j h_{ij} - h_{\text{min}}) \quad (8)$$

4) *Component Penalty* (F_{comp}): Because the objective is the reconstruction of a single vessel, the desired assembly is a single connected component. We apply a heavy penalty $\lambda_{\text{comp}} = 500.0$ for every disconnected sub-component C above the target of 1:

$$F_{\text{comp}} = \lambda_{\text{comp}} \cdot \max(0, |C| - 1) \quad (9)$$

This strongly drives the genetic algorithm toward a completely connected vessel.

G. Genetic Operators

1) *Selection*: Tournament selection is used with a tournament size of 2 to provide low selection pressure, promoting sustained genetic diversity across generations.

2) *Crossover*: We employ a Breadth-First Search (BFS)-Partition Crossover operator. To generate a child chromosome, the operator performs a BFS traversal over the active connection graph of the first parent, starting from a random shard. It inherits active match connections along the BFS tree until exactly half of the target edges are filled. The remaining edges required to form a complete spanning tree are then inherited directly from the second parent’s active connections. This ensures that the child inherits contiguous, topologically connected sub-assemblies from the parents.

3) *Mutation*: The algorithm utilizes several domain-aware mutation strategies executed probabilistically during reproduction. The base mutation rate p_m is set dynamically; it defaults to 0.15 but switches to a hyper-mutation rate of 0.40 if the population’s best fitness stagnates over successive generations, actively breaking the search out of local minima.

- **Neighborhood Mutate**: When triggered under the base mutation rate, this operator executes one of three local strategies with equal probability:

- 1) *Candidate Switch*: Selects an active pair group and swaps its current match hypothesis for another candidate within the same group, selecting the new hypothesis via rank-based sampling on density.
- 2) *Pair Switch*: Deactivates a randomly selected active edge, which temporarily breaks the spanning tree into two disconnected components. It then identifies all inactive candidate edges that bridge the two components and activates one (via rank-based sampling) to guarantee the assembly remains fully connected.
- 3) *Triangle-Builder*: Exploits the inherent geometric structure of pottery by actively searching for triplet pairs that mutually agree with each other. It takes an active pair of sherds and searches for a third sherd where the available pairwise match hypotheses form a geometrically consistent loop (i.e., their relative transforms compose correctly). When such an agreeing triplet is found, the operator ensures that two of these three matching pairs are activated in the assembly, solidifying a structurally validated triad core without violating the spanning tree constraint.

- **Macro Mutate (Hyper-mutation)**: If the algorithm enters the $p_m \geq 0.40$ hyper-mutation state, a massive, sherd-centric structural mutation occurs. A random target sherd is selected, and all of its active connections are completely severed. The algorithm then iterates through the fragmented components and systematically reconnects them using the highest-density available bridges, effectively forcing a structural reboot around the target sherd.

4) *Elitism*: The top $e = 6$ chromosomes are preserved directly into the next generation without modification, ensuring the best solution found is never lost.

IV. EXPERIMENTAL SETUP

A. Dataset

We evaluate on the SfS++ benchmark dataset [2], which contains 3D scanned pottery sherds from ten different archaeological pots (labeled A through J). Each sherd is represented as a point cloud with surface normals, breakline points, and rim segmentation. The ground truth graph G^* and transforms $\{T_i^*\}$ are used to compute accuracy metrics.

B. Evaluation Metrics

We report the standard SfS++ accuracy metrics:

- **Sherd Accuracy (SA)**: The percentage of correctly placed sherds over the total number of sherds. A sherd is considered correctly placed if at least one of its edges is placed correctly within a rotation threshold and translation threshold of the ground truth transform.
- **Edge Accuracy (EA)**: The percentage of correctly identified pairwise connections over the total number of ground truth connections across the entire assembly.

C. Baseline

We compare against the BAISER beam search of SfS++ [2] run on identical input (same pairwise matches after pruning). Since both methods share the feature extraction and matching stages, any performance difference is attributable solely to the search algorithm.

D. Hyperparameters

Table I lists all GA hyperparameters used in our experiments.

TABLE I
GA HYPERPARAMETERS

Parameter	Value	Rationale
Population size	1000	Supports large search spaces
Max generations	100	Extended convergence margin
Elitism count	10	Preserves top 1% solutions
Base Mutation (p_m)	0.15	Steady local exploration
Hyper-Mutation (p_m)	0.40	Escapes deep local minima
Tournament size	2	Low selection pressure
λ_{comp}	500.0	Strong component penalty
λ_{overlap}	4000.0	Severe interpenetration penalty
λ_{cons}	20.0	Consensus loop closure reward
τ_t	8.0 mm	Consensus translation tolerance
τ_R	0.22 rad	Consensus rotation tolerance
h_{min}	1%	Minimal overlap tolerance

E. Execution Environment

All experiments were run using the default parameters defined in the codebase, which uses deterministic random number generation (seeded with 42) to ensure exact reproducibility.

V. RESULTS

A. Reconstruction Accuracy

Table II compares our GA against baseline methods across the pottery datasets from the SfS++ benchmark. Our approach achieves results comparable to SfS++ on pots A and C (100% SA and EA), and nearly perfect accuracy on Pot F (missing only 2 edges). On more challenging pots like B, G, H, and J, the method lags behind SfS++, showing the feasibility of evolutionary search while identifying areas for improvement.

B. Fitness Convergence Analysis

Table III summarizes the fitness improvements achieved by the GA across all evaluated potteries. The algorithm optimizes the fitness function in every case, showing strong convergence properties regardless of whether the final assembly aligns with the archaeological ground truth.

For example, on Pot A, the best fitness improves dramatically from an initial $-111,533$ to a highly positive 34.6 . Even on heavily penalized datasets like Pot H and Pot J, the GA successfully reduces the massive overlap and cycle penalties by hundreds of thousands of points (e.g., Pot H improves from $-879,344$ to $-115,945$).

C. Fitness Component Breakdown

Analysis across multiple pots shows a three-stage optimization process in the GA convergence. In the initial phase (typically Gens 0–15), the algorithm focuses on pruning redundant or contradictory pairwise matches. Initially, the population contains high-density chromosomes with many active pairs (e.g., 23 for Pot A) that often include contradictory geometric constraints. These contradictions lead to large Edge Residual Penalties as the BFS-based assembly construction cannot satisfy the divergent transforms. As the GA learns to deactivate these problematic genes, reducing the active pairs to a consistent subset, the residual penalty drops significantly. Simultaneously, the Cycle Penalty is typically eliminated early (e.g., by Gen 14 for Pot A), establishing a topologically consistent search space.

In the middle phase (Gens 15–35+), the best fitness often enters a refinement plateau (e.g., near -517 for Pot A or $+28$ for Pot F), where the algorithm explores different match combinations while keeping residuals low. This corresponds to the discovery of plausible but partially incorrect assemblies. Finally, in the late phase, successful runs exhibit a breakout from this local optimum (occurring at Gen 35 for Pot A and as late as Gen 43 for Pot F), further reducing residuals and overlaps to reach the ground-truth reconstruction. Failure cases (e.g., Pot B) typically stall in the middle-phase plateau and fail to achieve this final breakout.

D. Structural Evolution

Structural metrics show the topological changes in the best-performing assembly over generations. Active pairs for individuals in the GA drop from 24 down to 10 to avoid contradictory pairwise matches which consequently gives a higher fitness reward.

E. Visual Reconstruction Progression

The algorithm traverses through plausible but incorrect false positive configurations before escaping the local optimum to reach the perfect ground-truth reconstruction.

F. Ablation Study

Table IV presents results from our fitness component ablation on Pot A, where each component is disabled individually while all others remain active.

The ablation results for Pot A show the relative importance of each fitness component. The Rotation Residual and Overlap Penalty emerged as the most critical factors; disabling them led to the largest drops in Edge Accuracy (-10 and -8 edges respectively) and a collapse in Sherd Accuracy to 62.5% . This shows that precise rotational alignment and physical non-overlap are the primary drivers of a globally consistent assembly.

Disabling the Cycle Penalty or the Inlier Reward also significantly degraded the reconstruction, reducing Edge Accuracy to 60.0% . This confirms that both high-quality local surface matches and global topological consistency are required to reach the ground-truth state. Interestingly, the Connectivity

TABLE II
RECONSTRUCTION ACCURACY ACROSS MULTIPLE POTTERIES

Method	Pot A (8)		Pot B (9)		Pot C (4)		Pot D (28)		Pot E (31)		Pot F (6)		Pot G (7)		Pot H (11)		Pot I (27)		Pot J (11)	
	SA	EA	SA	EA	SA	EA	SA	EA	SA	EA	SA	EA	SA	EA	SA	EA	SA	EA	SA	EA
Jigsaw	25.0	6.7	22.2	6.7	0.0	0.0	-	-	-	-	0.0	0.0	42.9	20.0	0.0	0.0	-	-	0.0	0.0
FRAISER	0.0	0.0	22.2	6.7	100.0	40.0	-	-	-	-	33.3	11.1	0.0	0.0	0.0	0.0	-	-	0.0	0.0
PuzzleFusion++	62.5	46.7	33.3	20.0	50.0	20.0	-	-	-	-	100.0	100.0	100.0	100.0	0.0	0.0	-	-	0.0	0.0
SfS	0.0	0.0	100.0	100.0	100.0	100.0	82.1	74.2	100.0	77.4	100.0	100.0	0.0	0.0	81.8	76.5	88.9	70.8	18.2	4.8
SfS++	100.0	100.0	100.0	100.0	100.0	100.0	82.1	83.3	100.0	100.0	100.0	100.0	100.0	100.0	90.9	88.2	92.6	90.8	72.7	76.2
GA (Ours)	100.0	100.0	100.0	100.0	100.0	100.0	57.1	40.9	80.6	64.5	100.0	100.0	100.0	100.0	27.3	17.6	85.2	80.0	-	-

SA: Sherd Accuracy (%); EA: Edge Accuracy (%). Number of sherds per pot indicated in parentheses.

TABLE III
FITNESS CONVERGENCE SUMMARY ACROSS ALL POTTERIES

Pottery	Gen 0 Best Fitness	Gen 49 Best Fitness
Pot A	-111,533.0	34.6
Pot B	-20,658.5	-4,064.9
Pot C	364.5	439.2
Pot F	-755.6	106.8
Pot G	-5,326.7	174.6
Pot H	-879,344.0	-115,945.0
Pot J	-316,476.0	-113,604.0

TABLE IV
FITNESS COMPONENT ABLATION STUDY (POT A)

Configuration	Sherd %	Edge %	Δ Edge
All components (full)	100.0	100.0	baseline
No inlier reward	100.0	60.0	-6
No connectivity reward	100.0	100.0	0
No cycle penalty	75.0	60.0	-6
No trans. residual	100.0	46.7	-8
No rot. residual	62.5	33.3	-10
No overlap penalty	62.5	46.7	-8
Inlier reward only	100.0	66.7	-5

Reward had no impact on Pot A’s final result, suggesting that for this specific geometry, the cycle and overlap constraints were sufficient to enforce a single connected component. Finally, the Inlier Reward Only configuration achieved only 66.7% EA, proving that while pairwise matching provides a strong starting point, global fitness optimization is essential to resolve the remaining combinatorial ambiguity.

VI. DISCUSSION

A. False Positives vs. Bad Reconstructions

A key challenge in designing the fitness function is distinguishing between a geometrically valid reconstruction and the true archaeological reconstruction. The fitness function evaluates local geometric consistency, such as match quality, cycle closure, edge residuals, and physical overlap, but lacks awareness of the expected global shape of the vessel.

In cases with lower Edge Accuracy such as Pot B, G, H, and J, the primary failure mode is not a visually coherent false positive, but rather bad reconstructions. In these cases, the algorithm places sherds in physically implausible or chaotic arrangements where sherds hover over each other, fail to connect cleanly, or overlap despite the penalty.

However, in other cases, the GA converges to an assembly that produces no overlap, no disagreements in residuals, and strong connectivity, yielding a high fitness score, but the shape is incorrect to a human observer expecting symmetry. We define these as false positives. For example, a false positive reconstruction of Pot F (using an alternative seed) shows sherds connected in a geometrically plausible way that is nevertheless incorrect.

This false positive phenomenon is also observable during the progression of a successful run. For instance, in the reconstruction of Pot A, the algorithm constructs a plausible but false positive partial assembly around generation 30, before successfully breaking out of the local optimum to find the true ground-truth configuration by generation 50. Since the fitness function does not encode any prior about the expected vessel profile or global symmetry, resolving false positives purely via pairwise constraints remains a limitation.

B. Overlap Penalty Effectiveness

The overlap penalty successfully reduces interpenetration artifacts in the decoded assembly. By precomputing voxel-downsampled collision clouds and using kd-tree nearest-neighbor queries, the algorithm achieves sufficient collision detection coverage at manageable computational cost. The penalty formulation efficiently bounds the penalty to avoid destabilizing the overall fitness score, while distinguishing legitimate edge contact from physical interpenetration.

C. Computational Cost

GA fitness evaluation involves BFS pose propagation, cycle penalty computation ($O(n^3)$ in sherd count), and overlap detection ($O(n^2|V|\log|V|)$), where $|V|$ is the number of points in the voxel-downsampled collision cloud. With $n = 4$ sherds, per-chromosome evaluation is extremely fast. With a population of 100 and 50 generations, total runtime remains highly competitive even on standard laptops. Though, for more complex pots like Pot D, the runtime is still too computationally expansive and need more computation to parallelize the GA even more.

D. Limitations and Future Work

One primary limitation of the current approach is that it treats reassembly as a single global optimization task, attempting to fit all sherds simultaneously. This can be difficult to achieve as the search space grows. Future work could explore

a hierarchical GA strategy: performing initial iterations with smaller populations and generations to find multiple optimal partial assemblies for individual sherds, and then using these as building blocks for subsequent GA rounds. This would combine the exploratory advantages of genetic search with a structured, step-by-step construction process.

Another significant limitation is the lack of a global vessel profile prior, which often leads to geometrically valid but archaeologically incorrect false positives. Future iterations of the GA should incorporate features that evaluate whether a proposed assembly is physically plausible according to the expected vessel shape. Integrating such symmetry and profile constraints would help the algorithm differentiate between the global ground truth and local geometric local optima.

VII. CONCLUSION

This paper presented a genetic algorithm-based system for the 3D reassembly of axially symmetric pottery. By replacing greedy search strategies with a population-based evolutionary approach, we successfully reconstructed two pots from the SfS++ benchmark and achieved results comparable to state-of-the-art beam search on Pot F. However, for the remaining complex vessels in the dataset, our current GA implementation lagged behind existing state-of-the-art methods. Our analysis revealed a characteristic three-stage optimization process, where the GA first prunes contradictory pairwise matches before entering a refinement plateau and finally breaking out into the ground-truth configuration.

Ablation studies confirmed that global geometric consistency, specifically enforced by rotation residual and overlap penalties, is essential for resolving the inherent ambiguity of featureless pottery fragments. While the system demonstrated strong performance on moderate datasets, challenges remain for complex vessels with high fragment counts. Future work will focus on implementing a hierarchical GA framework to build assemblies incrementally and integrating global vessel profile priors to eliminate false positive reconstructions. This work establishes evolutionary computation as a promising direction for solving large-scale combinatorial problems in archaeological restoration.

REFERENCES

- [1] J. Lu, Y. Liang, H. Han, J. Hua, J. Jiang, X. Li, and Q. Huang, "A survey on computational solutions for reconstructing complete objects by reassembling their fractured parts," 2025. [Online]. Available: <https://arxiv.org/abs/2410.14770>
- [2] S. J. Yoo, S. Liu, M. Z. Arshad, J. Kim, Y. M. Kim, Y. Aloimonos, C. Fermuller, K. Joo, J. Kim, and J. H. Hong, "Structure-from-sherds++: Robust incremental 3d reassembly of axially symmetric pots from unordered and mixed fragment collections," 2025. [Online]. Available: <https://arxiv.org/abs/2502.13986>
- [3] N. Rasheed and M. J. Nordin, "A survey of computer methods in reconstruction of 3d archaeological pottery objects," *International Journal of Advanced Research*, vol. 3, pp. 712–714, 01 2015.
- [4] Q.-X. Huang, S. Flory, N. Gelfand, M. Hofer, and H. Pottmann, "Reassembling fractured objects by geometric matching," *ACM Trans. Graph.*, vol. 25, no. 3, p. 569–578, Jul. 2006. [Online]. Available: <https://doi.org/10.1145/1141911.1141925>

- [5] S. Winkelbach and F. M. Wahl, "Pairwise matching of 3d fragments using cluster trees," *International Journal of Computer Vision*, vol. 78, no. 1, pp. 1–13, Jun. 2008. [Online]. Available: <https://link.springer.com/article/10.1007/s11263-007-0121-5>
- [6] G. Papaioannou, T. Schreck, A. Andreadis, P. Mavridis, R. Gregor, I. Sipiran, and K. Vardis, "From reassembly to object completion: A complete systems pipeline," *J. Comput. Cult. Herit.*, vol. 10, no. 2, Mar. 2017. [Online]. Available: <https://doi.org/10.1145/3009905>
- [7] R. Gregor, I. Sipiran, G. Papaioannou, T. Schreck, A. Andreadis, and P. Mavridis, "Towards automated 3d reconstruction of defective cultural heritage objects," in *Eurographics Workshop on Graphics and Cultural Heritage*, 2014. [Online]. Available: <https://api.semanticscholar.org/CorpusID:17749630>
- [8] R. Pintus, K. Pal, Y. Yang, T. Weyrich, E. Gobbetti, and H. Rushmeier, "A survey of geometric analysis in cultural heritage," *Computer Graphics Forum*, vol. 35, no. 1, pp. 4–31, 2016. [Online]. Available: <http://reality.cs.ucl.ac.uk/courses/gchstageomanalys2014/pintus2016survey-lowres.pdf>
- [9] M. F. C. Pinho, G. L. A. Mota, and G. A. O. P. da Costa, "A deep learning approach to assist in pottery reconstruction from its sherds," *Heritage*, vol. 8, no. 5, 2025. [Online]. Available: <https://www.mdpi.com/2571-9408/8/5/167>
- [10] D. Sholomon, O. David, and N. S. Netanyahu, "A genetic algorithm-based solver for very large jigsaw puzzles," in 2013 IEEE Conference on Computer Vision and Pattern Recognition. IEEE, Jun. 2013, p. 1767–1774. [Online]. Available: <http://dx.doi.org/10.1109/CVPR.2013.231>
- [11] D. Sholomon, O. E. David, and N. S. Netanyahu, "Genetic algorithm-based solver for very large multiple jigsaw puzzles of unknown dimensions and piece orientation," in *Proceedings of the 2014 Annual Conference on Genetic and Evolutionary Computation*, ser. GECCO '14. ACM, Jul. 2014, p. 1191–1198. [Online]. Available: <http://dx.doi.org/10.1145/2576768.2598289>
- [12] E. Sizikova and T. Funkhouser, "Wall painting reconstruction using a genetic algorithm," *J. Comput. Cult. Herit.*, vol. 11, no. 1, Dec. 2017. [Online]. Available: <https://doi.org/10.1145/3084547>

APPENDIX A

IN-DEPTH VISUAL RESULTS PER POTTERY

This appendix provides detailed visual documentation for each pottery assembly evaluated in our study. For each pottery (Pot A through Pot J), we present the fitness convergence curves, the per-component fitness breakdown, and a side-by-side comparison of our final GA reconstruction against the archaeological ground truth.

APPENDIX B

ABLATION STUDY VISUAL COMPARISON

This section provides a visual comparison of the final reconstructions for Pot A across all ablation configurations discussed in Section VI. These visuals illustrate how disabling specific fitness components leads to characteristic reconstruction failures, such as sherd interpenetration (no overlap penalty) or fragmented assemblies (no inlier reward).